40 When a radioactive isotope is set up close to a counter, a count rate of 38 000 counts / s is obtained. The table shows the count rate from the isotope over a three-year period.

time / years	count rate / counts / s
0	38 000
1	26 000
2	17 000
3	12 000

What is the half-life of the isotope?

A less than 1 year

B more than 1 year but less than 2 years

C more than 2 years but less than 3 years

D more than 3 years